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Food Rescue Connect: Smart Food Donation and Redistribution Management System

Industry Context

A CASE STUDY REPORT

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ABSTRACT

FoodRescue Connect is a proposed digital platform designed to reduce food wastage by connecting food donors with verified recipients and volunteers. Restaurants, hotels, households, event organizers, and other donors can list surplus food, while eligible recipient organizations can view available donations and request or reserve them. Volunteers may support pickup and delivery activities, and administrators can manage users, donations, requests, and system activities.

The purpose of this document is to design comprehensive test cases for FoodRescue Connect based on the assigned problem statement. The testing process focuses on verifying correctness, functionality, reliability, security, usability, and performance. Functional requirements such as registration, login, food donation creation, food search, request management, volunteer assignment, notifications, and administrative operations are considered.

The document applies several software test design techniques. Equivalence Partitioning is used to divide valid and invalid input values into classes. Boundary Value Analysis is used for fields such as quantity, contact information, and expiry-related constraints. Decision Tables are applied to donation acceptance and request approval conditions. State Transition Testing is used for donation and request status changes.

The resulting test suite provides traceable coverage of major requirements and includes normal, boundary, invalid, and exceptional conditions. Actual Result and Pass/Fail fields are intentionally provided as execution fields to be completed during practical testing.

1. INTRODUCTION

Software testing is a systematic process used to determine whether a software system satisfies its specified requirements and behaves as expected under different conditions. Testing is especially important for applications that coordinate real-world activities because failures can affect users, resources, schedules, and trust in the system.

FoodRescue Connect addresses the problem of surplus food going unused while people and organizations may require food support. The proposed platform creates a digital connection between donors, recipients, volunteers, and administrators. Because the system handles user accounts, food information, requests, pickup arrangements, and notifications, it requires comprehensive testing.

The assigned task requires the preparation of test cases covering functionality, reliability, usability, and correctness. Therefore, this document begins by identifying the requirements and test scenarios and then defines detailed test cases with test data and expected outcomes.

The test case design follows a black-box testing approach. The tester verifies the externally observable behavior of each feature without depending on the internal implementation. The test suite also uses structured techniques so that important input classes, boundary conditions, business rules, and state changes are not missed.

The report is intended to serve as a practical test case design document. During actual execution, the tester can enter the observed Actual Result and Pass/Fail Status for every test case.

2. REQUIREMENTS TO BE TESTED

2.1 Functional Requirements

- FR01 – User Registration: New users shall be able to create an account using required personal and contact details.
- FR02 – User Login and Logout: Registered users shall be able to securely log in and log out.
- FR03 – Role Management: The system shall distinguish donor, recipient, volunteer, and administrator roles.
- FR04 – Donation Creation: Donors shall be able to create a food donation with food type, quantity, location, preparation time, and expiry information.
- FR05 – Donation Validation: The system shall reject incomplete, invalid, expired, or logically inconsistent donation information.
- FR06 – Donation Search: Recipients shall be able to search and filter available donations.
- FR07 – Donation Request: Eligible recipients shall be able to request available donations.
- FR08 – Request Approval/Reject: Donors or authorized administrators shall be able to approve or reject requests according to business rules.
- FR09 – Volunteer Assignment: Available volunteers shall be assignable to approved pickup/delivery activities.
- FR10 – Status Tracking: Donation/request statuses shall be updated as activities progress.
- FR11 – Notifications: Relevant users shall receive notifications for important events.
- FR12 – Administration: Administrators shall be able to manage users, donations, requests, and reports.

2.2 Non-Functional Requirements

- NFR01 – Usability: Screens and messages should be understandable and easy to navigate.
- NFR02 – Performance: Common operations should respond within an acceptable time under normal load.
- NFR03 – Reliability: Transactions should complete correctly without data loss.
- NFR04 – Security: Unauthorized users should not access restricted functions or other users' sensitive information.
- NFR05 – Compatibility: The application should operate correctly on supported browsers and devices.

- NFR06 – Availability: Core services should remain available during expected operating periods.
- NFR07 – Data Integrity: Donation, request, user, and status information should remain accurate and consistent.

3. TEST SCENARIOS

The following scenarios represent the major workflows that need to be tested for the FoodRescue Connect system.

Scenario ID	Test Scenario	Coverage
TS01	User registration	Valid, duplicate, incomplete and invalid registration data
TS02	Authentication	Login with valid and invalid credentials; logout
TS03	Role authorization	Verify permissions for donor, recipient, volunteer and admin
TS04	Create donation	Create donation with valid food and location details
TS05	Donation validation	Check missing, invalid, boundary and expired values
TS06	Search and filter	Search donations by type, quantity, location and availability
TS07	Request donation	Recipient submits, edits or cancels a request
TS08	Approve/reject request	Verify business rules for approval and rejection
TS09	Volunteer workflow	Assignment, acceptance, pickup and delivery status
TS10	Notifications	Verify event-based notification generation
TS11	Administration	User, donation and request management
TS12	Security	Unauthorized access, session and input validation
TS13	Performance	Response under normal and increased load
TS14	Usability and compatibility	Navigation, messages, browser/device behavior

The scenarios are mapped to requirements so that each major functional area has at least one corresponding test path. Detailed test cases in the next section expand these scenarios into executable steps.

4. TEST DESIGN TECHNIQUES

4.1 Equivalence Partitioning

Equivalence Partitioning divides input data into classes where values are expected to produce similar system behavior. For example, a registration email can be divided into valid email addresses and invalid email formats. Testing one or more representative values from each class reduces unnecessary duplication while maintaining coverage.

4.2 Boundary Value Analysis

Boundary Value Analysis focuses on values at and around the limits of an input field. For example, if a donation quantity accepts 1 to 100 units, values 0, 1, 2, 99, 100, and 101 are useful test values. Boundary testing is particularly important for quantity, age, field length, dates, and time restrictions.

4.3 Decision Table Testing

Decision Table Testing is useful when the output depends on multiple conditions. In FoodRescue Connect, approval of a donation request may depend on donation availability, recipient eligibility, expiry status, and authorization. A decision table ensures combinations of these conditions are considered.

4.4 State Transition Testing

State Transition Testing verifies that an object moves through permitted states and prevents invalid transitions. A donation may move from Available to Requested, Approved, Picked Up, Delivered, or Cancelled depending on actions and conditions. Testing these transitions helps prevent inconsistent records.

4.5 Requirement Coverage

Each functional requirement is linked to one or more test cases. The tester can use the requirement identifiers to confirm that the final test execution covers the intended functionality.

5. DETAILED TEST CASES – USER AND DONATION MODULE

ID	Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC01	TS01 Registration	Open registration; enter all valid fields; submit	Name=Ravi; email=ravi@test.com; valid phone; valid password	Account created and confirmation shown	_____	Pass/Fail
TC02	TS01 Registration	Submit registration with missing mandatory field	Email omitted	Validation message shown; account not created	_____	Pass/Fail
TC03	TS01 Registration	Enter malformed email	ravi@	Invalid email message shown	_____	Pass/Fail
TC04	TS01 Registration	Register with existing email	Existing registered email	Duplicate-account message shown	_____	Pass/Fail
TC05	TS02 Login	Enter valid credentials and submit	Registered email + correct password	User is logged in and dashboard displayed	_____	Pass/Fail
TC06	TS02 Login	Enter wrong password	Registered email + wrong password	Login rejected with safe error message	_____	Pass/Fail
TC07	TS02 Login	Logout from dashboard	Authenticated session	Session ends and login page displayed	_____	Pass/Fail
TC08	TS03 Authorization	Recipient attempts donor-only function	Recipient account + create-donation URL	Access denied; no donation created	_____	Pass/Fail
TC09	TS04 Create donation	Create donation with valid values	Cooked food; quantity 20; valid location; future expiry	Donation saved with Available status	_____	Pass/Fail
TC10	TS05 Validation	Submit donation with quantity 0	Quantity=0	Validation error; record not saved	_____	Pass/Fail
TC11	TS05 Validation	Submit donation with negative quantity	Quantity=-5	Validation error; record not saved	_____	Pass/Fail
TC12	TS05 Validation	Submit donation with expired date/time	Expiry earlier than current time	Donation rejected or marked invalid according to rules	_____	Pass/Fail

6. DETAILED TEST CASES – SEARCH, REQUEST AND VOLUNTEER MODULE

ID	Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC13	TS06 Search	Search by food type	Food Type=Rice	Matching available rice donations displayed	_____	Pass/Fail
TC14	TS06 Search	Search with no matching item	Food Type=NonExistingItem	No-result message shown without error	_____	Pass/Fail
TC15	TS06 Filter	Apply location filter	Location=Nellore	Only matching-location donations shown	_____	Pass/Fail
TC16	TS07 Request	Request an available donation	Recipient eligible; donation available	Request created and status shown	_____	Pass/Fail
TC17	TS07 Request	Request unavailable donation	Donation already reserved	Request prevented with appropriate message	_____	Pass/Fail
TC18	TS07 Request	Cancel pending request	Existing pending request	Request becomes Cancelled and availability updates correctly	_____	Pass/Fail
TC19	TS08 Approval	Approve eligible request	Available donation + eligible recipient	Request becomes Approved; relevant users notified	_____	Pass/Fail
TC20	TS08 Approval	Reject unauthorized approval attempt	Recipient attempts approval	Action denied; request unchanged	_____	Pass/Fail
TC21	TS09 Volunteer	Assign available volunteer	Approved request + available volunteer	Volunteer assignment saved and notification sent	_____	Pass/Fail
TC22	TS09 Volunteer	Assign unavailable volunteer	Volunteer already assigned	System prevents conflicting assignment	_____	Pass/Fail
TC23	TS09 Pickup	Mark pickup completed	Assigned volunteer + approved request	Status changes to Picked Up	_____	Pass/Fail
TC24	TS09 Delivery	Complete delivery	Picked-up request	Status changes to Delivered; completion recorded	_____	Pass/Fail

7. DETAILED TEST CASES – SECURITY, NOTIFICATION AND EXCEPTIONAL CONDITIONS

ID	Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC25	TS10 Notification	Create donation and check recipient notification	Valid donation	Notification generated for relevant users	_____	Pass/Fail
TC26	TS10 Notification	Approve request and check donor/volunteer alerts	Approved request	Correct notification generated without duplicate alerts	_____	Pass/Fail
TC27	TS11 Admin	Admin views user list	Admin account	Authorized user list displayed	_____	Pass/Fail
TC28	TS11 Admin	Normal user opens admin function	Recipient account	Access denied	_____	Pass/Fail
TC29	TS12 Security	Access protected page after logout	Logged-out session + protected URL	User redirected to login	_____	Pass/Fail
TC30	TS12 Security	Enter script-like input in text field	<script>test</script>	Input safely handled; no script execution	_____	Pass/Fail
TC31	TS12 Security	Use another user's request ID	Valid account + чуж request ID	Unauthorized data/action blocked	_____	Pass/Fail
TC32	TS13 Performance	Search donations under normal load	Typical dataset and users	Response within defined performance target	_____	Pass/Fail
TC33	TS13 Performance	Submit donation during high load	Multiple concurrent users	Transaction completes without corruption or duplicate records	_____	Pass/Fail
TC34	TS14 Usability	Navigate registration to dashboard	Normal user workflow	Labels, controls and navigation are clear	_____	Pass/Fail
TC35	TS14 Compatibility	Open system on supported browsers	Chrome/Edge/Firefox or supported browsers	Core functions operate consistently	_____	Pass/Fail
TC36	TS07 Exception	Two recipients request the final available donation nearly simultaneously	Donation quantity sufficient for one request only	System prevents over-allocation and maintains one valid allocation	_____	Pass/Fail

8. DECISION TABLE AND STATE TRANSITION TESTING

8.1 Decision Table – Donation Request Approval

Rule	Availability	Recipient Eligibility	Expiry	Authorization	Expected Action
D1	Available? Yes	Eligible? Yes	Expired? No	Authorized? Yes	Approve
D2	Available? No	Eligible? Yes	Expired? No	Authorized? Yes	Reject – unavailable
D3	Available? Yes	Eligible? No	Expired? No	Authorized? Yes	Reject – ineligible
D4	Available? Yes	Eligible? Yes	Expired? Yes	Authorized? Yes	Reject – expired
D5	Available? Yes	Eligible? Yes	Expired? No	Authorized? No	Reject – unauthorized

8.2 State Transition – Donation Lifecycle

State ID	Current State	Event/Action	Expected Next State
S1	Draft	Valid donation submitted	Available
S2	Available	Recipient request accepted	Requested
S3	Requested	Authorized approval	Approved
S4	Approved	Volunteer completes pickup	Picked Up
S5	Picked Up	Delivery completed	Delivered
S6	Available	Donor/admin cancellation	Cancelled
S7	Requested	Request rejected/cancelled	Available or Cancelled

The state model prevents invalid sequences such as marking a donation as Delivered before it has been Picked Up. State transition tests should verify both valid and invalid transitions.

9. TEST COVERAGE, TEST DATA AND EXECUTION GUIDELINES

9.1 Requirement-to-Test Coverage

Requirement	Area	Test Cases	Coverage
FR01	User Registration	TC01–TC04	Covered
FR02	Authentication	TC05–TC07	Covered
FR03	Role Management	TC08, TC20, TC28	Covered
FR04	Donation Creation	TC09	Covered
FR05	Donation Validation	TC10–TC12	Covered
FR06	Donation Search	TC13–TC15	Covered
FR07	Donation Request	TC16–TC18, TC36	Covered
FR08	Approval/Rejection	TC19–TC20	Covered
FR09	Volunteer Assignment	TC21–TC24	Covered
FR10	Status Tracking	TC23–TC24 + state tests	Covered
FR11	Notifications	TC25–TC26	Covered
FR12	Administration	TC27–TC28	Covered
NFR01–NFR07	Usability, performance, security, compatibility, integrity	TC29–TC35	Covered

9.2 Test Data Categories

- Normal data: Valid user details, valid donation details, valid requests, and ordinary workflow values.
- Boundary data: Minimum and maximum permitted quantities, field-length limits, and dates/times near expiry.
- Invalid data: Incorrect formats, missing fields, negative quantities, duplicate accounts, and invalid identifiers.
- Exceptional data: Concurrent requests, unavailable volunteers, expired donations, network/service interruptions, and unauthorized actions.
- Security data: Script-like input, protected URLs, invalid sessions, and attempts to access another user's records.

9.3 Execution Guidelines

Before execution, the tester should prepare the application environment, database, test accounts for each role, representative donation records, and supported browsers/devices.

10. TEST SUMMARY AND CONCLUSION

The FoodRescue Connect test case design provides a structured approach to validating a food donation and redistribution platform. The proposed test suite covers the major user roles and business workflows, including registration, authentication, donation creation, searching, requesting, approval, volunteer assignment, pickup, delivery, notifications, and administration.

The test cases deliberately include normal, boundary, invalid, and exceptional conditions. This helps identify defects that may not appear during simple happy-path testing. For example, testing zero and negative quantities validates input constraints, while testing simultaneous requests helps detect over-allocation and data consistency problems.

The use of Equivalence Partitioning reduces redundant test data while ensuring representative valid and invalid classes are tested. Boundary Value Analysis focuses on values close to limits. Decision Table Testing validates combinations of business rules, and State Transition Testing verifies that donation and request statuses follow valid workflows.

Non-functional aspects are also included. Security tests verify authorization and session protection, performance tests consider response behavior under load, usability tests examine navigation and clarity, and compatibility tests check operation across supported browsers. These tests help ensure that the system is not only functionally correct but also dependable and usable.

The document can be extended during actual implementation by adding defect IDs, execution dates, tester names, severity, priority, screenshots, and evidence for failed cases. Regression testing should be repeated after defects are corrected to ensure that changes have not introduced new problems.

In conclusion, comprehensive test case design is essential for a system such as FoodRescue Connect because it coordinates multiple users and time-sensitive food resources. The test cases presented in this report provide a practical foundation for validating the proposed system and achieving adequate requirement and functional coverage.